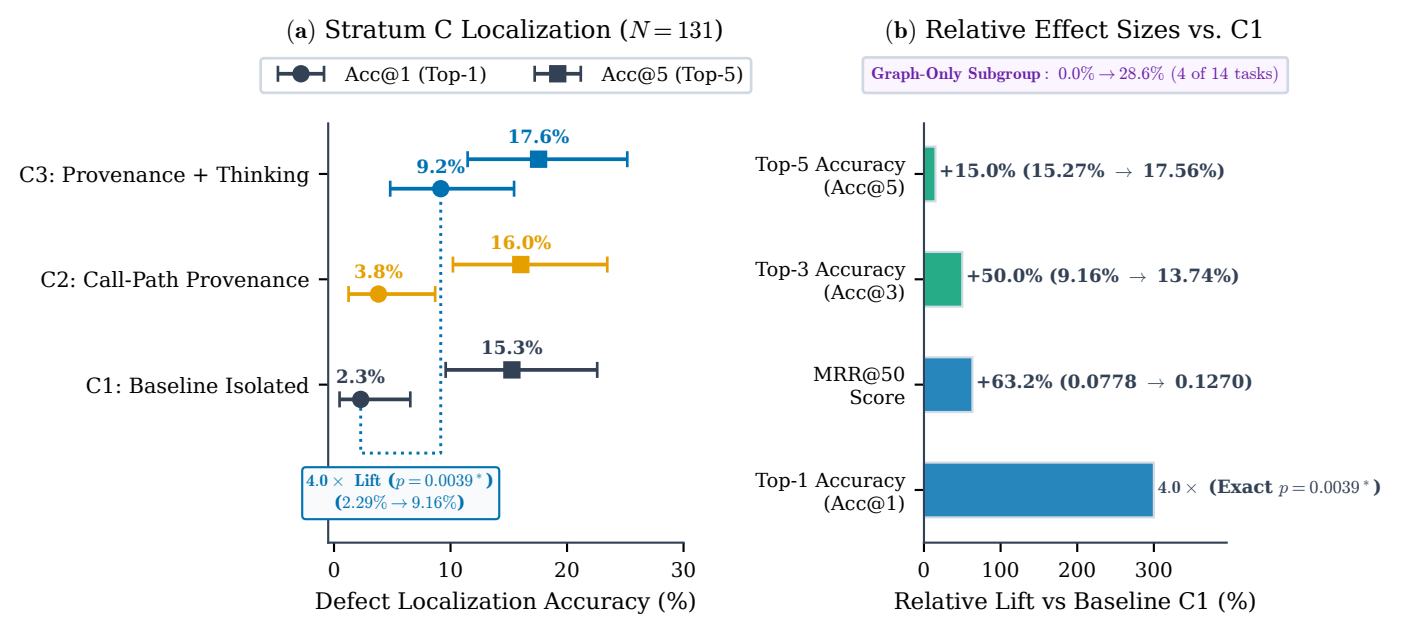
(a) Stratum C Localization ($N = 131$)
Acc@1 (Top-1)
Acc@5 (Top-5)
C3: Provenance + Thinking
C2: Call-Path Provenance
C1: Baseline Isolated
17.6%
9.2%
16.0%
3.8%
15.3%
2.3%
4.0× Lift ($p = 0.0039^*$)
(2.29% → 9.16%)
0
10
20
30
Defect Localization Accuracy (%)
(b) Relative Effect Sizes vs. C1
Graph-Only Subgroup: 0.0% → 28.6% (4 of 14 tasks)
Top-5 Accuracy (Acc@5)
+15.0% (15.27% → 17.56%)
Top-3 Accuracy (Acc@3)
+50.0% (9.16% → 13.74%)
MRR@50 Score
+63.2% (0.0778 → 0.1270)
Top-1 Accuracy (Acc@1)
4.0× (Exact $p = 0.0039^*$)
0
100
200
300
Relative Lift vs Baseline C1 (%)